The Zollman Effect

2026-03-30

Models of Scientific Communities

The plan for the rest of term is to look at formal models of scientific communities.

Really our priority is looking at the spread of knowledge across any community. But looking at **scientific** communities in particular is useful for two reasons.

1. It's arguable that they are models for everyone. Maybe, as the psychologist Alison Gopnik argues, we're all scientists from a very *very* young age.
2. For scientists, it's conceptually easy to separate out the decisions of (a) what to study and (b) what to believe about different areas of study.

Science is social. Scientists share evidence with one another, read each other's papers, update their beliefs and practices based on what colleagues find.

This seems like it should be good. More information should help people converge on the truth faster.

Today we'll look at a model that challenges this assumption.

Question 1: Is it always better for the scientific enterprise to have information shared as widely as possible.

We'll look at a mathematical model that suggests the answer is **no**.

Question 2: Is it good for scientists to be as rational as possible?

Again, in special circumstances, it seems the answer might be **no**. Having some **irrationally stubborn** people in the community might be good for the health of the community.

Setting up the Model

Imagine a community of medical researchers working on a disease. There's a **current treatment** that is well understood and a **new treatment** that might be better.

Each scientist has a credence (a probability between 0 and 1) that the new treatment is superior.

Scientists who think the new treatment is probably better will experiment on it. Those who think the old treatment is probably better will stick with the old one.

Why This Structure?

Two features of this setup matter.

First, scientists are **active investigators**, not passive observers. They choose what to work on based on what they currently believe.

Second, the old treatment is well understood, so experimenting on it generates no new information. Only experiments on the new treatment are informative.

If everyone stops working on the new treatment, the community stops learning about it permanently.

A Toy Example

Suppose four researchers, A, B, C, and D, assign the following probabilities to the new treatment being better:

A: 0.33, B: 0.49, C: 0.51, D: 0.66

A and B think the old treatment is probably better, so they work on it. C and D think the new treatment is probably better, so they test it.

Now suppose the new treatment really is better, but C and D both get unlucky results. Their experiments happen to suggest the new treatment is worse.

After Sharing Results

C and D share their (misleading) results with A and B. Everyone updates using Bayes's rule.

All four scientists now have lower credences in the new treatment. Even C and D, who were mildly optimistic, now think the old treatment is better.

So everyone switches to the old treatment. Nobody is testing the new one anymore. And since the old treatment generates no new information, nobody will ever switch back.

The community has lost a better treatment forever, because of a run of bad experimental luck.

What Went Wrong

The misleading evidence was a statistical fluke. With enough further experimentation, C and D would have corrected the error.

But because all the evidence was pooled immediately, the fluke swept through the entire group at once. There was no one left still testing the new treatment who could have generated the corrective evidence.

Had D been unaware of C's results, she would have kept testing. Her credence would have stayed above 0.5, and eventually the truth would have come out.

Two Questions for You

1. Is this at all realistic?
2. If so, can you think of real-life cases that might look like this?

The Formal Model

The model behind this comes from two economists, Venkatesh Bala and Sanjeev Goyal.

There are two states of the world (the new treatment is better, or it isn't) and two actions (work on the new treatment, or stick with the old one).

The old treatment has a known expected payoff in both states. The new treatment has a lower expected payoff in one state and higher in the other.

Figure 1: Woman playing slot machine

The structure here is similar to what have become known as **multi-arm bandit** problems.

Imagine a slot-machine, like the one pictured, with a choice of arms, each with different payoff probabilities.

These problems typically have the following characteristics.

1. The payoffs are simple; often either just win or lose on each round.
2. Each arm has a different probability of winning.
3. The players have a rational prior probability of what each of those probabilities are.
4. Also, these probabilities can drift over time, in a random way. The player knows the rule for the random drift, but not how any arm has drifted.
5. If the player chooses an option, they observe whether they win or lose, but not whether they would have won/lost if they'd chosen another option.

Three somewhat interesting facts (or at least interesting to me) about these problems.

1. They typically have no known **solution**, in the sense that a mathematician can look at the rules of the game and tell you what the optimal strategy is.
2. In practice, what we find by trial and error is that the best strategy is wide exploration at first, followed by settling on a favored option, but still occasionally exploring other options.
3. There are a lot of real life situations that feel like they have this structure - think about choosing a coffee shop or a route for walking home.

They are relevant to science in a straightforward way.

- Experiments are difficult, and cost time and money.
- You don't know what you would have got from doing some other experiment.
- It's possible that always doing the experiment with the high expected value, like always choosing the arm with the highest win probability, is **not** the optimal long-run strategy.

Each round, every scientist performs an experiment on whichever treatment she currently favors. In bandit terms, scientists always **exploit** rather than **explore**.

She gets a payoff drawn from a distribution that depends on which state the world is actually in. Higher payoffs are more likely when you're testing the better treatment.

She then observes her own result **and the results of some of her neighbors**, and updates her beliefs using Bayes's rule.

The word "some" is doing a lot of work there. Which neighbors she can see depends on the **network structure**.

Network Structure

Place the scientists on a graph. Each scientist can observe the experimental results of scientists she is directly connected to, and only those scientists.

Zollman (2007) compares three network structures for communities of the same size:

Cycle: scientists arranged in a circle, each seeing only her two immediate neighbors.

Wheel: like a cycle, but one scientist (the "hub") is connected to everyone else.

Complete graph: every scientist sees every other scientist's results.

The Natural Expectation

The complete graph gives every scientist access to all available evidence. The cycle gives each scientist access to very little.

You might expect that more information is always better, and that the complete graph would outperform the cycle.

The opposite turns out to be true.

The Simulation Results

Reliability

Zollman runs 10,000 simulations for each network type, in a world where the new treatment really is better. He then checks how often the community converges on the correct answer.

The **cycle** reaches the correct consensus most often. The **wheel** does somewhat worse. The **complete graph** does worst of all.

Communities with less information flow are more reliable.

There is a trade-off, though.

Among the communities that do reach the right answer, the **complete graph** gets there fastest. The cycle is slowest.

So the community with the most communication is fastest but least reliable. The one with the least communication is slowest but most reliable.

The mechanism is the one we saw in the toy example with A, B, C and D, but it generalizes.

In the complete graph, a streak of bad luck from one scientist's experiments spreads immediately to everyone. If the unlucky results push enough scientists below 0.5, the whole community abandons the new treatment at once.

In the cycle, bad results stay local. They might convince a few nearby scientists to stop testing, but scientists on the other side of the cycle never see those results. They keep testing, and eventually their data corrects the error.

Inertia

Sparsely connected networks have more “inertia” in two senses.

First, fewer scientists change their behavior after any single round of results, because fewer scientists are exposed to any given piece of evidence.

Second, the last remaining scientist testing the new treatment is harder to dislodge in a sparse network. In a complete graph, if there’s only one scientist left testing the new approach, she’s seeing negative results from everyone else. In a cycle, she’s only seeing results from two neighbors.

Preserving Diversity

The key insight is about **diversity of practice**. A well-functioning scientific community needs some scientists testing each option, at least until the evidence becomes overwhelming.

In less connected networks, misleading evidence stays contained in one part of the community while other parts continue exploring. This preserves the diversity of approaches long enough for the truth to emerge.

In highly connected networks, everyone reacts to the same evidence simultaneously, so the community can swing together in the wrong direction.

The Speed-Accuracy Trade-Off

Zollman checks all 112 distinct networks of six scientists. Across all of them, the relationship holds: networks that are more accurate tend to be slower, and networks that are faster tend to be less accurate.

The five most accurate networks are all minimally connected (you can't remove an edge without splitting the graph in two). The five fastest are all close to complete graphs.

When Does This Matter?

The trade-off isn't always a problem.

If you're confident the community is close to the truth already, more communication is fine. The risk of a misleading streak overwhelming everyone is low, and you benefit from the speed.

If the stakes of getting the wrong answer are high and you want to make sure the community gets it right, less communication is better. You sacrifice speed for reliability.

Whether speed or accuracy matters more depends on the specific case.

Implications

The Zollman Effect

The finding that less communication can improve collective outcomes in epistemic communities has come to be called the **Zollman effect**.

It shows up beyond the Bala-Goyal model. Similar results appear in models where agents search for the best option among several possibilities: too much connectivity leads the group to settle on the first good option found, rather than exploring to find a better one.

Zollman connects this to the history of peptic ulcer disease. For decades, the medical community believed ulcers were caused by excess stomach acid. A competing theory held that they were caused by a bacterium.

One prominent paper finding no bacteria in human stomachs was enough to push the field away from the bacterial theory. The acid theory dominated for almost fifty years before the bacterial theory was vindicated.

If parts of the community had been less aware of that one paper, some researchers might have kept testing the bacterial hypothesis, and the field might have converged on the correct answer sooner.

The Independence Thesis

Mayo-Wilson, Zollman, and Danks (2011) draw a broader conclusion from results like the Zollman effect.

They argue for what they call the **Independence Thesis**: methodological prescriptions for individual scientists and methodological prescriptions for scientific communities are logically independent of each other.

A method that works well for an individual scientist working alone may fail when used by a group. And a collection of methods that works well for a group may include methods that would fail for an isolated individual.

Consider a “decreasing epsilon greedy” strategy. This is a pretty natural solution to bandit problems; you explore less and less as time goes on.

The scientist mostly tests whichever theory has performed best so far, but occasionally tests alternatives. The rate of exploration decreases over time.

Alone, this converges on the right answer. But in a group, it can fail: the agents converge on each other’s behavior too quickly, and everyone locks in on a suboptimal theory before gathering enough evidence.

Good for Groups, Bad for Individuals

Consider a group where each scientist has a "favorite theory." She mostly tests her favorite, but occasionally tests others. Individually, any one of these scientists might get stuck on the wrong theory forever.

But as a group, they guarantee that every theory gets tested persistently. The best theory eventually produces enough good results that it propagates through the network, and the whole community converges on it.

The second result is the more surprising one. The group succeeds precisely because its members are not individually optimal.

Each scientist's stubbornness about her favorite theory looks like a bias. But that bias is what keeps the community exploring all the options. If every scientist had instead used the individually optimal strategy, they might all have converged prematurely on the same (possibly wrong) answer.

This echoes a theme from twentieth century philosophy of science: dogmatic commitment to a theory, which seems irrational for an individual, can benefit a scientific community by preserving the diversity of approaches.

The Independence Thesis means we can't just "scale up" from individual epistemology to social epistemology.

If you design a methodology by asking "what should an isolated rational scientist do?" and then tell every scientist in a community to follow that advice, you may get worse outcomes than if you had designed the methodology with the community in mind.

This has implications both for normative epistemology (what methods should we recommend?) and for the history of science (how should we evaluate scientific communities that tolerated seemingly irrational individuals?).

For Next Time

We'll look at how robust the Zollman effect is, what happens when diversity of belief leads to polarization, and how outside actors can manipulate epistemic networks.

Read O'Connor, Chapter 4, sections 4.2-4.4, and Rosenstock et al. (2017).